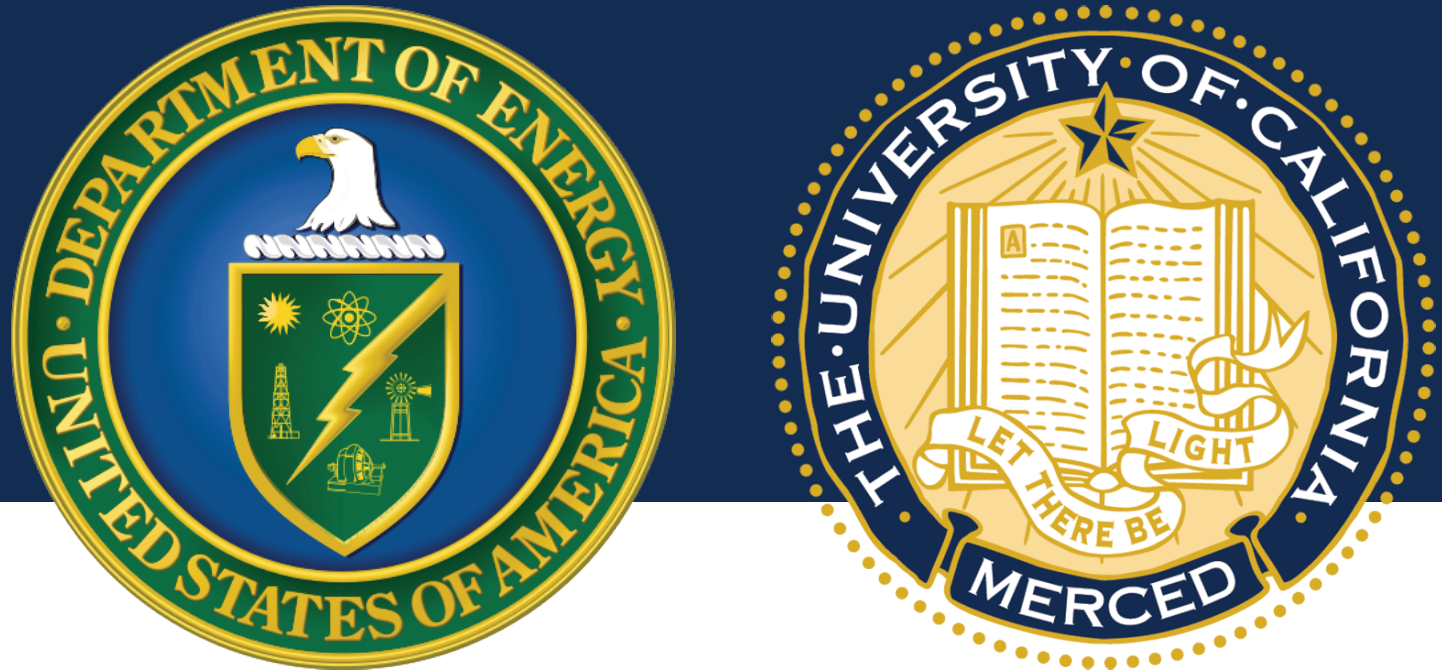


Different low-level warm-cloud responses to dust-cloud vertical configurations over land and ocean from ARM observations

Guoqing Gong¹ and Adeyemi Adebisi^{1,*} Email: ggong2@ucmerced.edu
¹ Department of Life and Environmental Sciences, University of California, Merced, USA



Motivation

- Mineral dust dominates aerosol mass but its cloud impacts remain highly uncertain.
- Dust-cloud interactions depend strongly on the relative vertical position.
- The same dust-cloud configuration can produce contrasting cloud responses over ocean and land.
- Long-term observations resolving dust vertical structure are needed to constrain warm-cloud.

Scientific Question

- How does dust vertical structure differ between a marine and continental environment?
- How do low-level warm-cloud macro- and microphysical properties respond to dust loading under different configurations?
- How do meteorological regimes modulate these dust-cloud sensitivities?

Data and Methods

Atmospheric Radiation Measurement



Sites: ENA and SGP.
Period: Jan 1st 2016 to Dec 31st 2023.
Variables: extinction coefficient, depolarization ratio, scattering ratio, LWP, cloud layer heights and meteorological variables.

Dust Identification and dust property calculation

Depolarization ratio corrected with scattering ratio $\rightarrow$ PDR
 $t = S + S\delta_m - \delta_m$
 $x_{num} = t\delta_{aerosol} - \delta_m$
 $x_{den} = S - 1 + S\delta_m - \delta_{aerosol}$
 $PDR = \frac{x_{num}}{x_{den}}$

$$f_{dust} = \begin{cases} 1, & PDR \geq \delta_d \\ 0, & PDR \leq \delta_{nd} \\ \frac{(PDR - \delta_{nd})(1 + \delta_d)}{(\delta_d - \delta_{nd})(1 + PDR)}, & \delta_{nd} < PDR < \delta_d \end{cases}$$

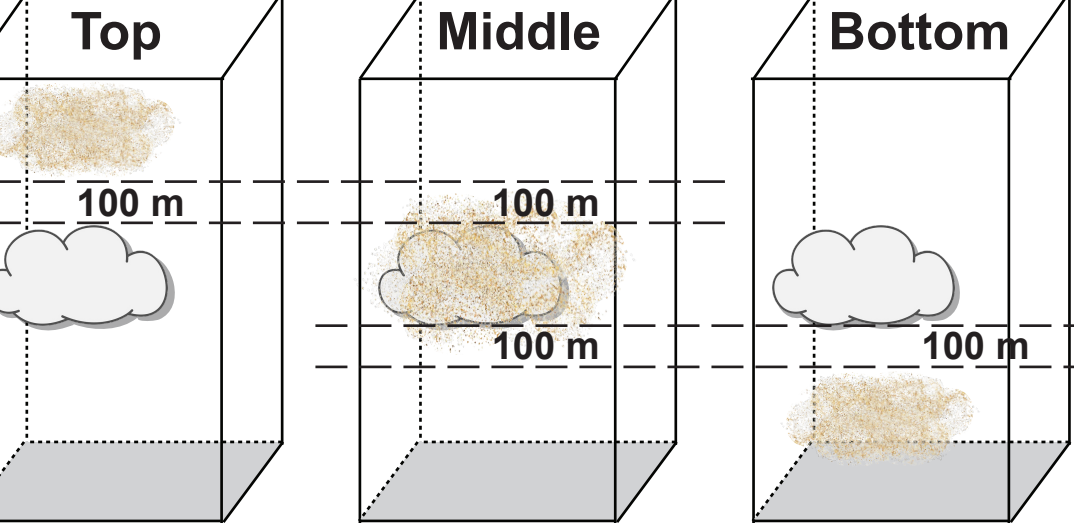
$ext_{dust}(z) = f_{dust}(z)ext_{aerosol}(z)$
 $DOD = \int_{Z_{base}}^{Z_{top}} ext_{dust}(z)dz$

Low level warm-cloud Identification and cloud property calculation

Single layer, liquid phase
Cloud top height < 3 km
Temperature > 0 °C

$$COD = \int_{Z_{base}}^{Z_{top}} ext_{cloud}(z)dz$$
$$R_e = \frac{2\rho_l}{3} \frac{LWP}{COD}$$

Configuration classification



Top: dust base $\geq$ cloud top+100 m.
Middle: cloud overlapping dust layer (± 100 m band).
Bottom: dust top $\leq$ cloud base-100 m and meteorological variables.

Figure 2. Concept illustration of the three dust-cloud configurations.

Aerosol cloud interaction: log-log regression

$$ACI_{cloud\ property} = \frac{\partial \ln(\text{cloud property})}{\partial \ln(DOD)}$$
$$ACI_{cloud\ property}^{(LWP)} = \left. \frac{\partial \ln(\text{cloud property})}{\partial \ln(DOD)} \right|_{LWP}$$

LWP bins (0-20, 20-50, >50 g m⁻²)

$$ACI_{cloud\ property}^{(var)} = \left. \frac{\partial \ln(\text{cloud property})}{\partial \ln(DOD)} \right|_{var}$$

RH, Temp, LTS: median threshold; VV: threshold at 0
Layer definitions: RH/Temp/VV averaged 750-850 hPa;
LTS = 0700 - 01000

Results

Result 1: Dust-Cloud Configuration Statistics

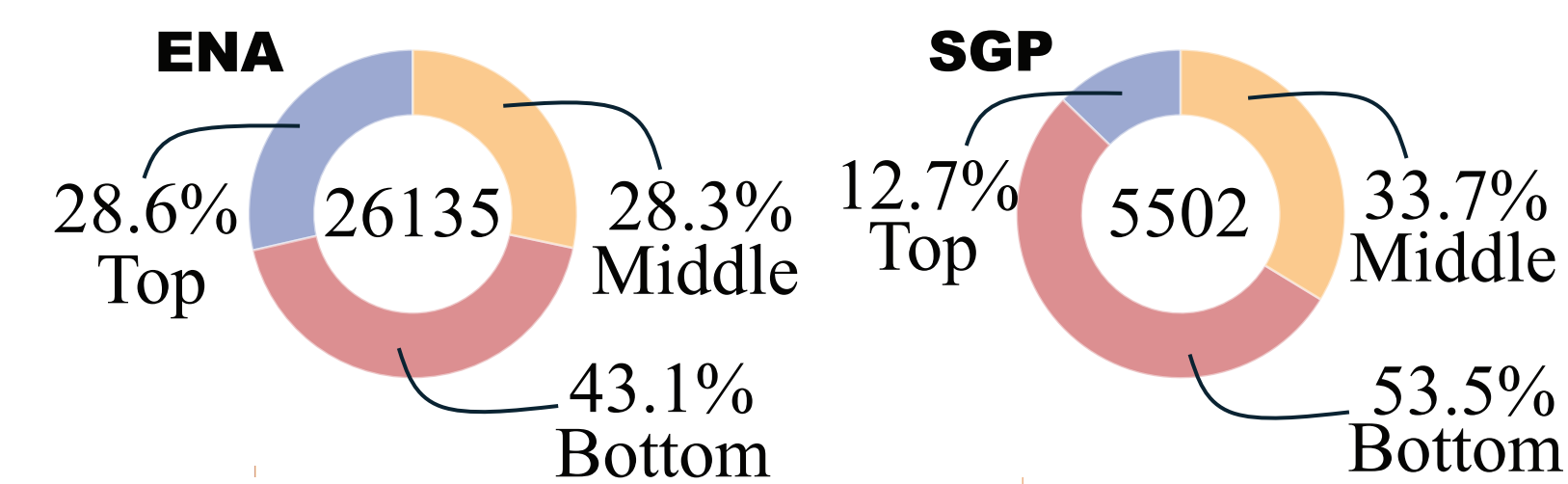


Figure 3. Classification of dust-cloud vertical categories.

Result 2: Configuration- and LWP-dependent Responses

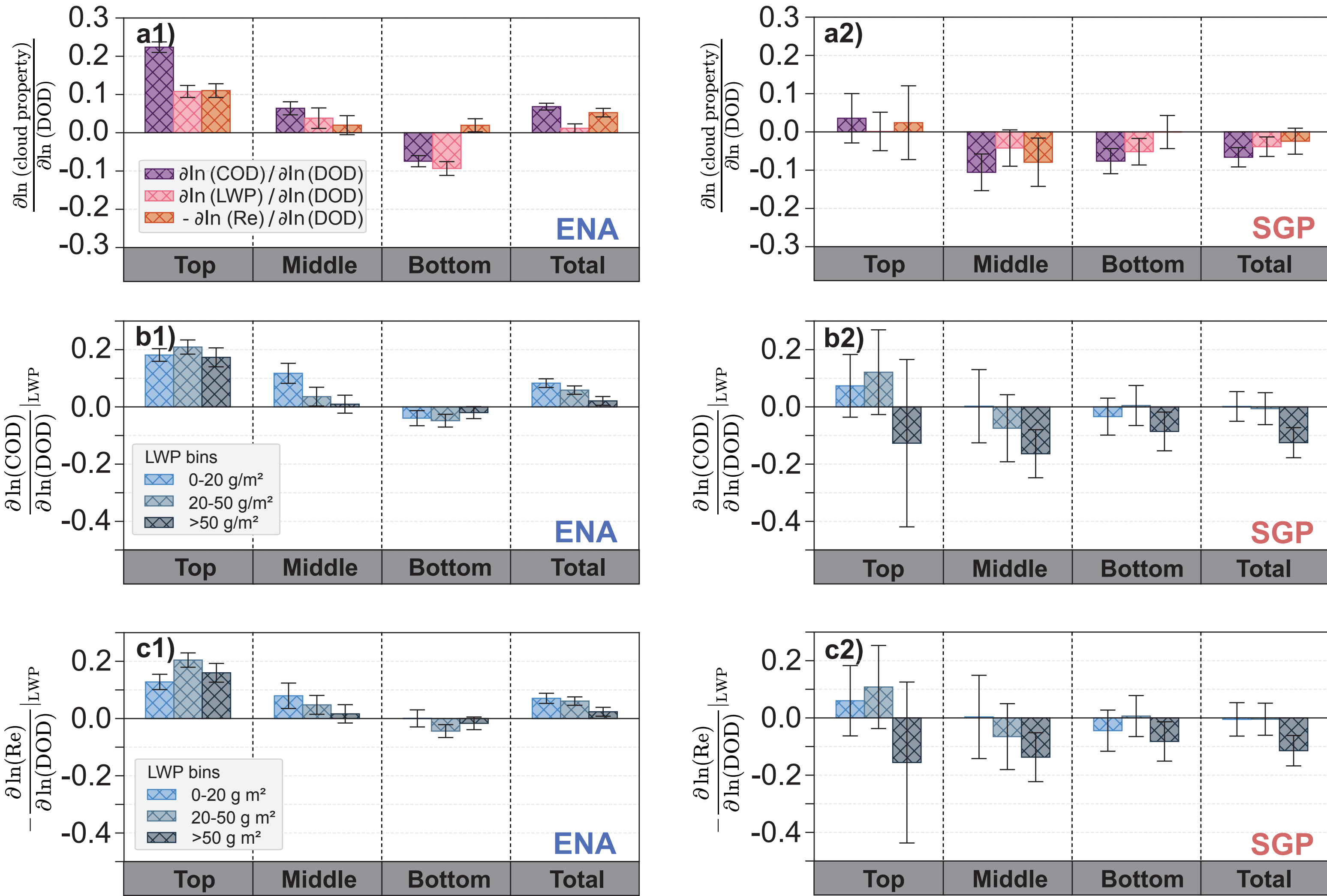


Figure 4. (a1) Cloud property sensitivity at ENA. (b1) Aerosol-cloud interaction with respect to COD ($\partial \ln(COD)/\partial \ln(DOD)$) and (c1) Re ($-\partial \ln(Re)/\partial \ln(DOD)$), evaluated under three LWP regimes. Vertical lines denote the 95% confidence intervals. Panels (a2)-(c2) show the corresponding results for SGP.

- ENA:** Top and Middle configurations exhibit cloud brightening, which weakens with increasing LWP, while the Bottom configuration remains characterized by cloud weakening.
- SGP:** Cloud responses are predominantly negative, with increasing LWP amplifying cloud weakening and promoting more frequent sensitivity sign reversals across configurations.

Conclusion

- Cloud responses to dust are primarily controlled by relative vertical position rather than dust loading alone.
- Marine and continental environments exhibit distinct configuration-dependent cloud responses.
- Meteorological regime and cloud state further modulate these responses, leading to greater sensitivity and sign reversals over land than over ocean.

Result 3: Meteorological modulation of cloud response

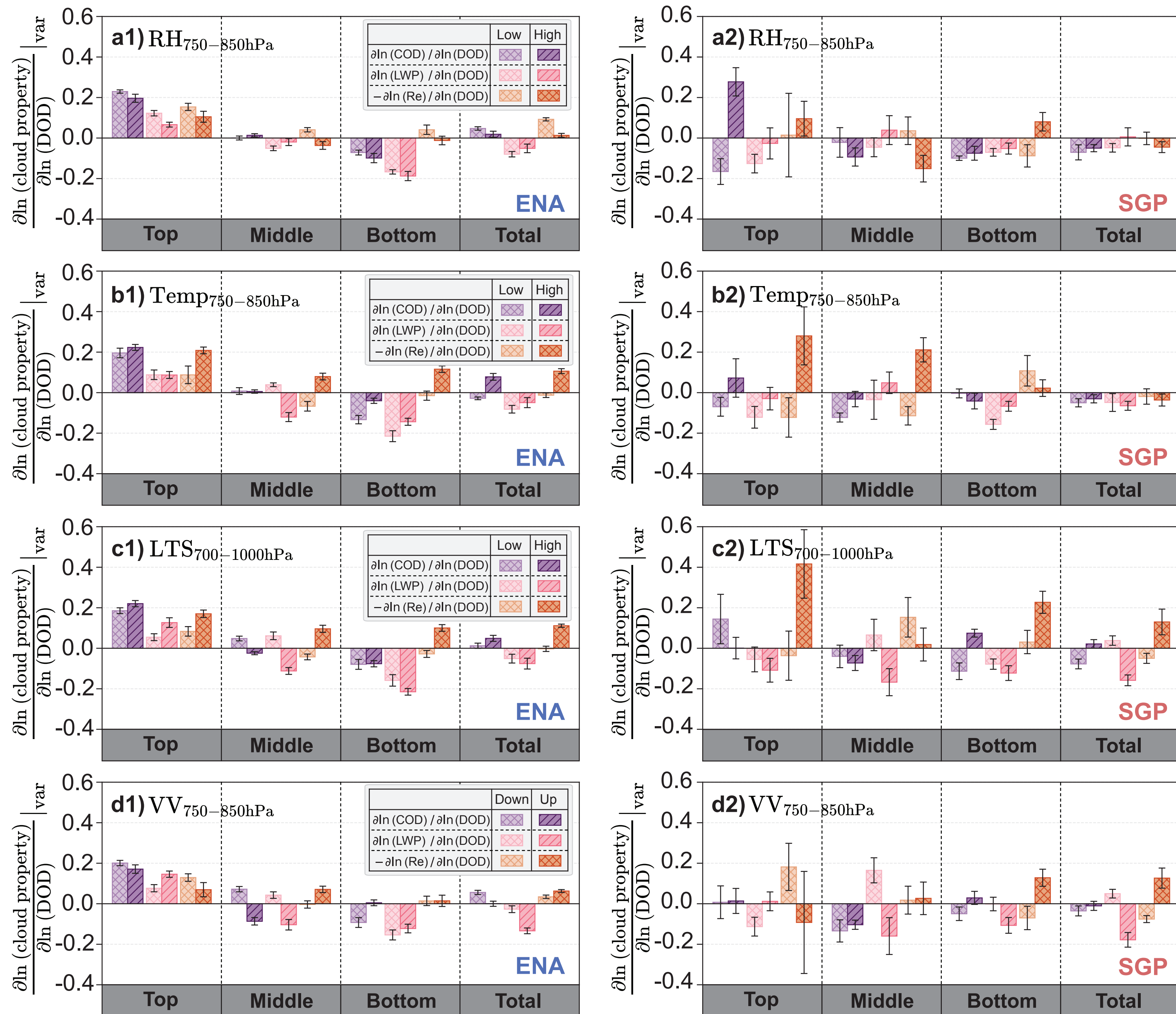


Figure 5. Responses of cloud properties to DOD under different meteorological regimes. Panels show results for (a1) relative humidity, (b1) temperature, (c1) lower-tropospheric stability, and (d1) vertical velocity, with each variable separated into low and high regimes. Panels (a2)-(d2) show the corresponding results for SGP.

- ENA:** meteorology mainly changes magnitude while preserving sign pattern.
- SGP:** meteorology frequently triggers sign reversals.
- Cloud responses in continental environments are more sensitive to meteorological variability than in marine environments.

Reference

Kok, Jasper F., et al. "Mineral dust aerosol impacts on global climate and climate change." Nature Reviews Earth & Environment 4.2 (2023): 71-86.
Liang, Yongen, et al. "Distinct impacts of aerosol size on aerosol-cloud interactions over ocean and land regions in Eastern China." Journal of Geophysical Research: Atmospheres 131.2 (2026): e2025JD045635.